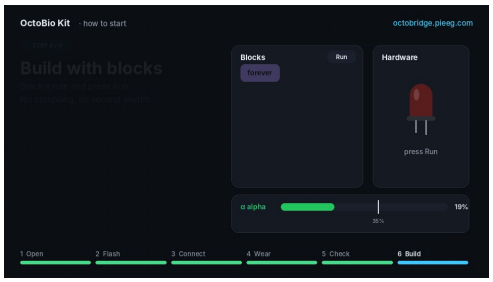
SOFTWARE & PROTOCOL GUIDE · SKU PIA24028P

octoBridge, Blocks, Serial & Bluetooth

How the kit is programmed: the octoBridge pages and block levels, what the EEG readings mean, the text protocol between octoBridge and the starter board, the Octopus 16 Bluetooth packet, the JavaScript SDK and the Arduino path.



2	115200	51 B	5	0
Devices live at once	Baud, line protocol	BLE packet / sample	Block levels	Compiling needed

01 OCTOBRIDGE PAGES

Page	What it does
Home	What you can read, what you can move, ready mixes
Flash	Firmware for both boards — first time only
Connect	Pair the Octopus 16 (Bluetooth) and/or the starter board (USB serial)
Board	Live numbers, lights, motor and screen
Blocks	Stack a program and run it live
Learn	What a reading is allowed to mean
Write	The same program written like a page of text
Repeat	Six ready machines to run and remix

02 BLOCK LEVELS AND READINGS

Level	Name	Audience
1	Story	Primary school
2	Words	
3	Bands δ θ α β γ	
4	Measure (numbers, units)	
5	Raw values	University

Reading	Definition
Band power	δ 0.5–4 · θ 4–8 · α 8–13 · β 13–30 · γ 30+ Hz, relative
Focus	β / (β + θ), uncalibrated
Relax	α / (α + β), uncalibrated
RMS	Signal size in µV
EEG ok	Green when RMS is 2–200 µV

Every lesson is one condition and one action. Thresholds in the lessons (50 lux, 300, 35 %) are starting values — check your own readings on the Board page.

03 STARTER BOARD SERIAL PROTOCOL

The OctoBio Kit firmware talks plain text over USB at **115200 baud**, one line per message: **KIND key=value key=value**. octoBridge waits for **INF protocol=1 ready** after the port opens, then sends **STREAM 250**.

```
DAT pot=512 lux=310 dist_cm=42.0 temp_c=23.40 hum=41.2 moisture=620 sound=160 button=0 pir=1 ir=- ax ay az
```

One text line per reading, fields "key=value" separated by spaces, ending in a newline. About 4 lines per second after STREAM 250.

For research and education only. Not a medical device. Specifications subject to change.

Board → browser	Meaning
`INF protocol=1 ready`	Firmware ready
`DAT …`	Sensor readings
`OK cmd=1`	Command accepted
`EVT …`	Event, e.g. IR key
`ERR …`	Command rejected

Browser → board	Effect
`STREAM 250` · `STREAM 0`	Readings every 250 ms · stop
`LED ON | OFF | FLASH`	Red LED D10
`LED2 ON | OFF`	Light D8
`SERVO 90`	Servo angle 0–180°
`BUZZER ON | OFF` · `RELAY ON | OFF`	Buzzer · relay
`LCD text` · `LCD2 text`	LCD line 1 · line 2

DAT fields: pot, lux, dist_cm (negative = missed echo), temp_c, hum, moisture, sound, button, pir, ir, ax/ay/az. Protocol as used by Octo Studio in the OctoBio-kit repository (protocol version 1).

04 OCTOPUS 16 BLUETOOTH

Name	`Octopus-16-XXXX`
Service	`4fafc201-1fb5-459e-8fcc-c5c9c331914b`
Data (notify)	`beb5483e-36e1-4688-b7f5-ea07361b26a8`
Command (write)	`c0de0001-36e1-4688-b7f5-ea07361b26a8` — register bridge

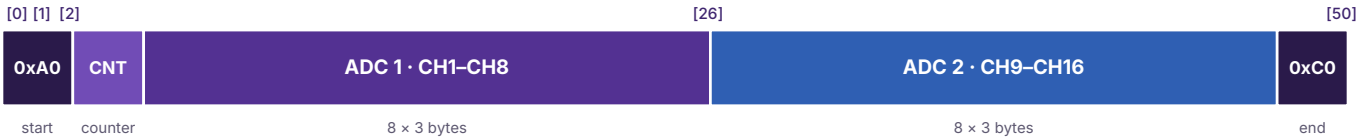


Figure 2 — 51 bytes per sample: 0xA0, counter, 16 channels × 3 bytes (24-bit big-endian signed), 0xC0. µV = raw × 1.2 / 4 / 8 388 607 × 10^6 (≈ 0.0358 µV per count).

05 YOUR OWN WEB APP AND ARDUINO CODE

JAVASCRIPT SDK · OCTOPUS 16 IN YOUR PAGE

```
<script src="https://cdn.jsdelivr.net/gh/pieeg-club/PiEEG-server@main/pieeg.js"></script>
<button id="connect">Connect</button>
<script>
  const eeg = new PiEEG();
  eeg.onBandPowers(p => console.log('alpha', p.Alpha, 'focus', eeg.getFocusIndex()));
  document.getElementById('connect').onclick = () =>
    eeg.connectBLE({ device: 'octopus-16' });      // must be called from a click
</script>
```

The starter board is a standard Uno-compatible board, so older students can write the same rule in C++ with the Arduino IDE using the pin table in OB-PIN-004. Uploading a sketch replaces the OctoBio firmware — open **Flash** in octoBridge and click **Flash octoBio Kit** to restore it (about 20 s). Supported: official Uno (16U2) or CH340 boards; not Leonardo, Mega or ESP32 boards.

06 RELATED DOCUMENTS

OB-DS-001 Product Datasheet	**OB-MD-002** Mechanical & Layout Drawing	**OB-PCB-003** Hardware & Manufacturing Specification
OB-PIN-004 Pinout & Connection Guide	**OB-QSG-005** Quick Start & Safety Guide	